

National University of Singapore
School of Computing

Semester 1, AY2026-27

CS4246/CS5446

Reinforcement Learning and Sequential Decision Making

Issued: 26 Aug 2026

Tutorial 5 Week 7: Mid-Term Review

Guidelines

You may discuss the content of the questions with your classmates. But everyone should work on and be ready to present ALL the solutions. [Note: Materials in this Tutorial cover only some of the topics relevant to the Mid-Term Exam on 7 October 2026.](#)

Problem 1: Classical Planning Review

Modified from [RN 10.3 Kindle Edition]

$Init(At(C_1, SFO) \wedge At(C_2, JFK) \wedge At(P_1, SFO) \wedge At(P_2, JFK)$
 $\wedge Cargo(C_1) \wedge Cargo(C_2) \wedge Plane(P_1) \wedge Plane(P_2)$
 $\wedge Airport(JFK) \wedge Airport(SFO))$
 $Goal(At(C_1, JFK) \wedge At(C_2, SFO))$
 $Action(Load(c, p, a),$
 PRECOND: $At(c, a) \wedge At(p, a) \wedge Cargo(c) \wedge Plane(p) \wedge Airport(a)$
 EFFECT: $\neg At(c, a) \wedge In(c, p)$
 $Action(Unload(c, p, a),$
 PRECOND: $In(c, p) \wedge At(p, a) \wedge Cargo(c) \wedge Plane(p) \wedge Airport(a)$
 EFFECT: $At(c, a) \wedge \neg In(c, p)$
 $Action(Fly(p, from, to),$
 PRECOND: $At(p, from) \wedge Plane(p) \wedge Airport(from) \wedge Airport(to)$
 EFFECT: $\neg At(p, from) \wedge At(p, to)$

- a) Given the action schemas and initial state from the figure above, what are all the applicable concrete instances of $Fly(p, from, to)$ in the state described by

$At(P_1, JFK) \wedge At(P_2, SFO) \wedge Plane(P_1) \wedge Plane(P_2) \wedge Airport(JFK) \wedge Airport(SFO)?$

- b) What is the result of executing the action $Load(C_2, P_2, JFK)$ from the initial state?

Problem2: Optimistic and Pessimistic Reachable Sets

Suppose the optimistic reachable set of a high-level plan is a superset of the goal set. Can we conclude anything about whether the plan will achieve the goal? What if the pessimistic reachable set does not intersect with the goal set? Explain.

Problem 3: Preference Modelling

Alex is given the choice between two games:

- **Game 1:** a fair coin is flipped, and if it comes up heads, Alex receives \$100. If the coin comes up tails, Alex receives nothing.
- **Game 2:** a fair coin is flipped twice. Each time the coin comes up heads, Alex receives \$50, and Alex receives nothing for each coin flip that comes up tails.

Alex prefers Game 2 to Game 1. After that, Alex is given two choices: Game 1 and directly receive \$50. Argue that Alex would prefer to receive \$50 compared to being allowed to participate in Game 1.

Problem 4: Decision Theory

A seasoned investor, Alice, has a utility function $U(x) = \ln(x)$, where x is total wealth, and has a choice between the following two alternatives:

- A: Win \$10,000 with probability 0.2
Win \$1,000 with probability 0.8
- B: Win \$3,000 with probability 0.9
Lose \$2,000 with probability 0.1

- a) If Alice current wealth is \$2,500, should she choose A or B ?
- b) If Alice current wealth is \$5,000, should she choose A or B ?
- c) If Alice current wealth is \$10,000, should she choose A or B ?
- d) Do you think this pattern of choices between A and B is reasonable? Why or why not?

Problem 5: Markov Decision Process

Consider a robot navigating a 10×10 grid world. At each time step spent in a non-goal cell, the robot incurs a cost of 0.05. The cell (10, 10) is the goal state and provides a reward of +100. The game will end whenever the robot reaches the goal cell. The robot may select one of four actions: UP, DOWN, LEFT, or RIGHT. Given a selected action, the robot moves in the intended direction with probability 0.8, or in either of the two directions orthogonal to the intended direction with probability 0.1 each. The duration of each movement is stochastic. Suppose the robot selects an action at time t :

- With probability 0.5, the movement takes one time step and is completed at time $t + 1$.
- With probability 0.5, the movement is delayed by one additional time step. In this case, the robot remains in its current cell at time $t + 1$, and the movement is completed at time $t + 2$.

The robot may select a new action only after its previously selected action has been completed. Thus, if an action is still in progress at time $t + 1$, the robot must wait and cannot select another action at that time. Its next decision is made only when the pending movement has finished. Model this situation as a Markov Decision Process. Clearly specify the state space, action space, transition probabilities, and reward function. Ensure that the state representation accounts for whether the robot has an unfinished action. *Optional*: Determine the optimal policy at time $t = 2$, and state the method used to obtain it.

Problem 6: Passive Reinforcement Learning

Consider a simple environment with three states:

$$S_0 \text{ (start)}, \quad S_1 \text{ (intermediate)}, \quad S_2 \text{ (terminal)}$$

A fixed policy π always takes the agent from $S_0 \rightarrow S_1$, then from $S_1 \rightarrow S_2$. Once in S_2 , the episode ends. The rewards are:

$$R(S_0 \rightarrow S_1) = 0, \quad R(S_1 \rightarrow S_2) = 1.$$

You want to learn the value function $U^\pi(s)$ for $s \in \{S_0, S_1, S_2\}$. Conceptually, explain how each of the following methods would update its value estimates based on experience:

(a) Monte Carlo Learning (MC)

- When does it update the value of each state?
- What data does it wait for before making an update?

(b) Temporal Difference (TD) learning, specifically $TD(\lambda = 0)$

- When and how does it update the value of each state?

- Why is it sometimes called a “bootstrapping” method?

(c) Adaptive Dynamic Programming (ADP)

- What additional information does it need to perform value updates?
 - How does it typically solve for (or iteratively refine) the value function?
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